

KRUGER MPUMALANGA INTL, South AfricaWMO: **682904**Lat: **25.388S**Lon: **31.099E**Elev: **862**StdP: **91.39**Time Zone: **2.00 (E02)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	8.8	-8.9	1.9	20.3	-6.2	2.5	19.6	7.2	18.7	6.4	18.4	1.9	260	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	33.0	18.2	31.2	18.2	30.0	18.2	22.6	27.7	22.0	27.1	21.4	26.1	4.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.6	24.3	20.4	16.8	23.6	20.0	16.4	23.3	71.5	27.6	68.8	27.4	66.4	26.2	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.1	5.5	4.6	36.4	1.3	1.9	3.7	37.8	2.9	38.9	2.2	40.0	1.2	41.4		
(5)			1.5	23.7	0.9	0.8	0.9	24.3	0.4	24.8	-0.1	25.2	-0.7	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	22.4	22.5	21.9	19.6	18.2	16.0	15.4	17.1	19.5	20.0	20.8	22.0
	DBStd	3.56	2.02	1.76	2.01	2.38	2.24	2.27	2.48	2.92	3.52	3.49	3.18	2.87
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	348	1	1	2	14	30	77	96	59	29	21	13	5
	CDD10.0	3506	384	351	368	289	253	179	168	221	286	311	323	371
	CDD18.3	812	127	118	112	54	25	6	6	22	64	73	86	118
	CDH23.3	5532	682	606	599	269	178	79	79	256	677	684	635	788
CDH26.7	1578	167	130	141	48	27	8	10	64	249	269	217	248	
Wind	WSAvg	2.5	2.7	2.5	2.3	2.1	2.1	2.2	2.3	2.6	3.0	2.9	2.8	2.6
Precipitation	PrecAvg	804	140	133	109	50	14	9	8	10	18	62	121	127
	PrecMax	1537	281	608	215	108	70	45	48	30	84	110	216	235
	PrecMin	358	18	38	35	0	0	0	0	0	0	21	31	31
	PrecStd	271	72	143	55	23	16	11	11	9	20	27	43	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.0	31.9	30.1	29.8	27.9	28.1	31.0	34.0	35.1	34.1	34.1
		MCWB	20.4	20.8	20.0	18.9	14.7	13.0	12.3	14.3	15.0	17.1	18.4	21.1
	2%	DB	30.0	29.9	29.9	28.0	27.0	25.8	25.9	28.9	31.8	32.1	31.2	31.8
		MCWB	20.2	20.5	19.3	18.1	15.0	12.3	11.8	14.0	15.1	17.4	19.0	20.8
	5%	DB	28.2	28.1	28.1	26.2	25.1	23.8	23.2	26.2	29.9	29.2	29.0	29.2
		MCWB	20.5	20.2	19.4	18.1	14.9	11.8	11.4	13.2	14.9	16.9	18.9	20.7
	10%	DB	27.0	27.0	26.9	24.8	23.7	21.9	21.2	24.1	27.8	27.1	27.0	27.8
		MCWB	20.1	20.0	19.4	17.7	14.5	11.9	11.4	12.7	14.8	16.7	18.9	20.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	23.1	22.7	21.6	19.0	15.2	14.9	17.1	18.7	20.9	22.2	23.2
		MCDB	28.7	28.5	27.1	25.3	24.7	21.6	21.6	24.0	26.4	29.6	27.9	29.4
	2%	WB	22.7	22.0	21.7	20.6	17.6	14.3	13.9	16.0	17.6	19.3	21.2	22.2
		MCDB	27.1	27.6	26.5	24.4	22.2	20.1	19.4	23.2	24.7	26.1	26.4	28.1
	5%	WB	21.9	21.3	20.8	19.8	16.8	13.7	13.2	15.1	16.7	18.5	20.4	21.5
		MCDB	26.2	26.0	25.3	23.5	21.5	19.4	18.7	21.7	24.1	25.6	25.6	27.0
	10%	WB	21.2	20.7	20.1	18.9	16.0	13.0	12.6	14.2	16.0	17.7	19.7	20.7
		MCDB	24.8	24.8	24.6	22.6	20.8	18.8	18.1	20.6	23.5	24.3	24.8	25.8
Mean Daily Temperature Range	5% DB	MDBR	8.6	8.6	9.2	9.0	10.7	11.6	11.3	11.9	11.8	10.4	9.1	8.8
		MCDBR	11.0	10.8	11.7	11.8	13.0	14.3	14.6	15.8	16.1	15.4	13.7	12.6
		MCWBR	3.9	3.8	4.1	5.2	5.2	5.7	5.7	6.0	5.5	4.8	4.8	4.8
	5% WB	MCDBR	8.9	9.4	9.5	9.1	10.6	11.4	11.3	12.7	12.8	12.4	10.7	10.6
		MCWBR	4.0	4.0	4.0	4.6	4.7	5.2	5.2	5.6	5.2	4.5	4.6	4.8
Clear-Sky Solar Irradiance	taub	0.401	0.393	0.388	0.371	0.336	0.320	0.321	0.381	0.432	0.451	0.419	0.418	
	taud	2.368	2.393	2.394	2.400	2.457	2.478	2.461	2.290	2.178	2.175	2.297	2.311	
	Ebn at Noon	943	936	910	879	871	866	878	855	854	871	920	929	
	Edh at Noon	132	126	119	109	93	87	92	119	146	155	141	140	
All-Sky Solar Radiation	RadAvg	6.04	5.97	5.29	4.51	4.17	3.77	4.00	4.63	5.23	5.31	5.68	5.97	
	RadStd	0.45	0.52	0.41	0.31	0.19	0.23	0.14	0.28	0.32	0.36	0.47	0.46	